

Flat Butterfly (FlatFly) Network-on-Chip Topology

Flat Butterfly (FlatFly) is a low-latency Network-on-Chip (NoC) topology designed to reduce the number of router traversals required for communication. Unlike a conventional mesh, where packets travel through many intermediate routers, a Flat Butterfly directly connects routers within the same row and column. This significantly reduces the maximum hop count while providing multiple shortest paths between source and destination.

Architecture

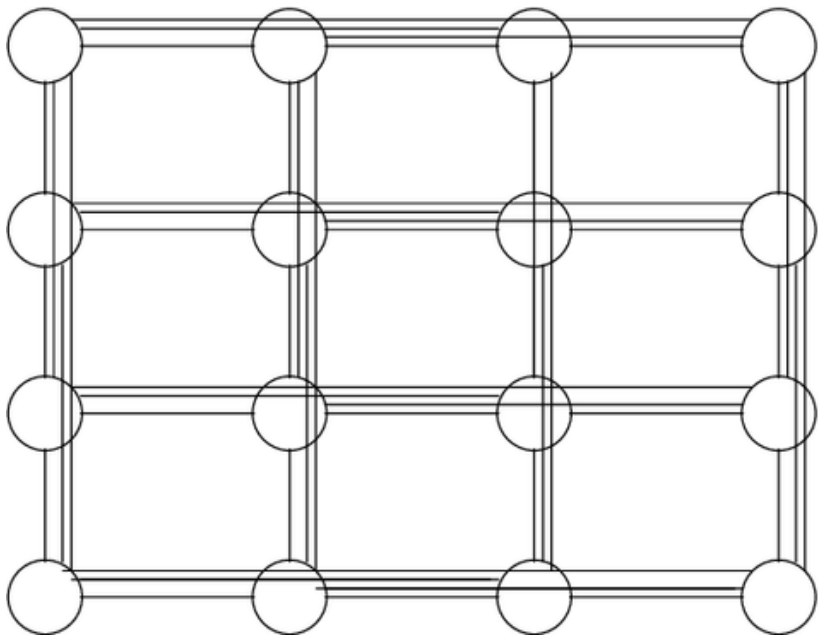
In a 2D Flat Butterfly with parameters $k \times k$, routers are arranged in a grid. Each router is connected to:

- All other routers in its row.
- All other routers in its column.
- Multiple processing elements (CPUs/IPs).

For example, with $k = 4$, each router has:

- 1.6 network ports (3 row links + 3 column links) $\{2(k-1)\}$
- 2.4 CPU ports (for $x_r = 2, y_r = 2$)
- 3.10 total ports

Packets typically reach any destination in at most two hops, making FlatFly one of the lowest-latency NoC topologies. example, if packet wants to go to (2,3) from (0,0) , it goes to (0,3) in 1st hop and then (2,3) in 2nd hop .(row first) similarly column first is also there.



Routing

Since multiple shortest paths exist between two routers (row-first or column-first), routing algorithms such as Random Minimal (ran_min) or adaptive routing distribute traffic across different paths to reduce congestion while maintaining the minimum hop count.

Advantage	Reason	Disadvantage	Reason
Low latency	Maximum of two router hops.	High router complexity	Router radix increases with network size.
Low hop count	Fewer router traversals reduce delay.	Large crossbar	Crossbar grows with the square of port count.
High throughput	Packets finish quickly, freeing network resources.	Complex arbitration	More ports require larger allocators.
Multiple shortest paths	Enables effective load balancing.	Long physical links	Direct row/column connections span large
Reduced congestion	Traffic can be distributed across alternative routes.	Higher area and power	More buffers, ports, and switching logic.
Suitable for AI/HPC	Excellent for communication-intensive	Limited scalability	Router complexity grows rapidly as k increases.

Conclusion

Flat Butterfly achieves very low communication latency by reducing the number of router traversals. However, this benefit comes at the cost of larger and more complex routers, making it most suitable for medium-sized high-performance systems rather than extremely large networks.

FatTree Network-on-Chip Topology

FatTree is a hierarchical Network-on-Chip topology designed to provide high scalability and high bandwidth while avoiding the bottlenecks present in conventional tree networks. Instead of relying on a single root router, FatTree uses multiple core routers and multiple parallel communication paths, allowing traffic to be evenly distributed throughout the network.

Architecture

A FatTree consists of three logical levels:

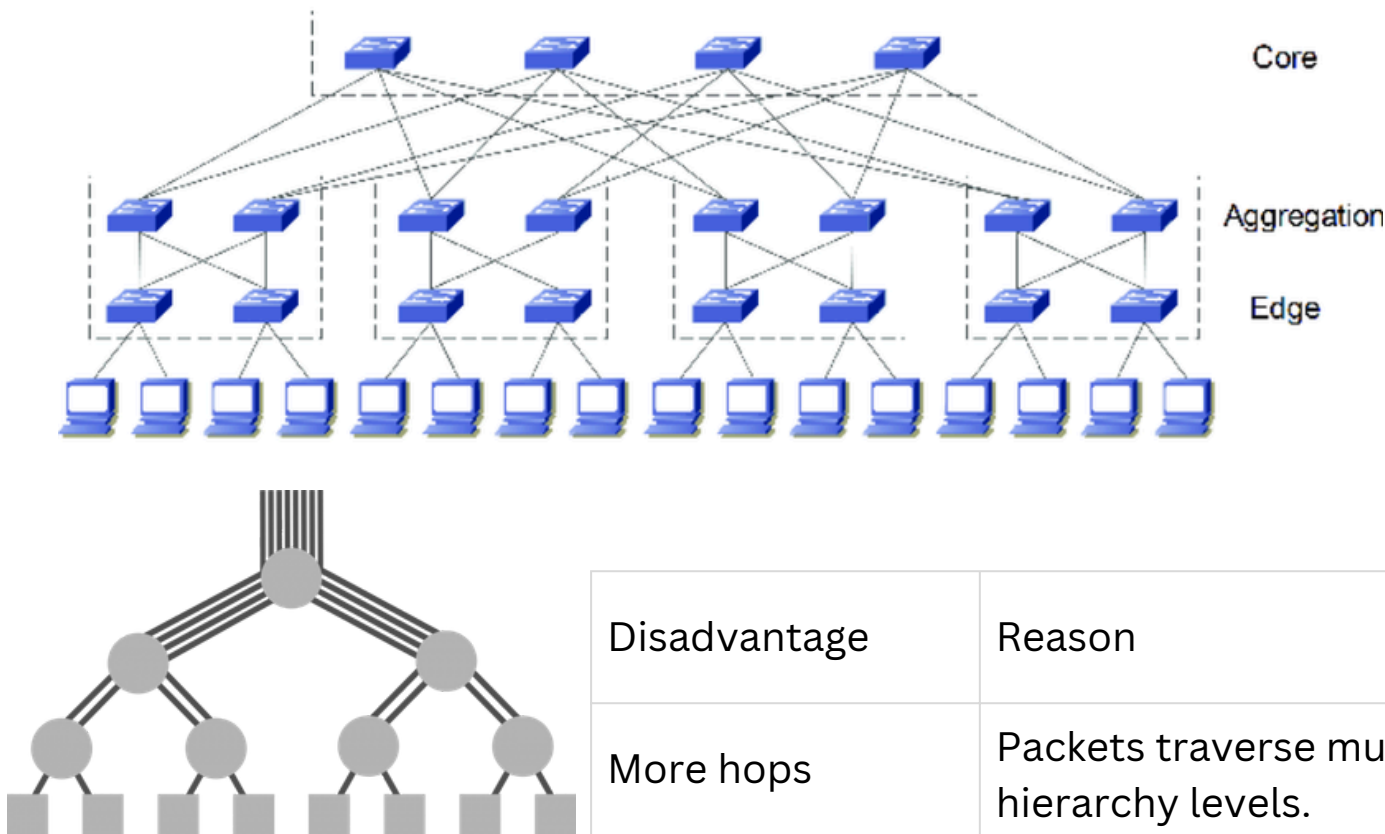
1. Leaf Routers: Connected directly to processing elements (CPUs/IPs).
2. Aggregation Routers: Connect groups of leaf routers.
3. Core Routers: Provide connectivity between different parts of the network.

Unlike a traditional tree, multiple core routers exist, and each aggregation router is connected to multiple core routers, providing several equal-cost paths between source and destination.

Only leaf routers connect directly to CPUs, while aggregation and core routers perform packet forwarding.

Routing

FatTree commonly employs Nearest Common Ancestor (NCA) routing. A packet travels upward only until it reaches the nearest router shared by both source and destination, then travels downward toward the destination. Since several equivalent upward paths are available, adaptive routing can choose the least congested path to improve network utilization.



Advantage	Reason
Highly scalable	Supports very large networks efficiently.
Small router complexity	Routers require relatively few ports.
High bisection bandwidth	Multiple parallel paths reduce bottlenecks.
Multiple equal-cost paths	Enables adaptive load balancing.
Fault tolerant	Alternative paths exist if one path becomes unavailable.
Better congestion handling	Routing can avoid busy links dynamically.
Lower router area	Smaller crossbars and arbitration logic.

Disadvantage	Reason
More hops	Packets traverse multiple hierarchy levels.
Higher latency	More router traversals than FlatFly.
Larger network	Requires many routers and interconnecting links.
Complex routing	NCA and adaptive routing are more sophisticated.
Increased implementation cost	More routers and wiring increase hardware cost.
Uneven utilization possible	Poor routing decisions may overload certain paths.

Conclusion

FatTree provides excellent scalability and bandwidth by distributing traffic across multiple core routers and multiple equal-cost paths. Although packets travel through more routers than in Flat Butterfly, the simpler router design and superior scalability make FatTree an ideal choice for very large NoCs and modern data-center interconnects.